EXPLORATORY DATA ANALYSIS ON A CARS DATASET

UNLOX Assignment

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Dataset: Vehicle Dataset from CarDekho

Dataset Source:
https://www.kaggle.com/datasets/nehalbirla/vehicle-dataset-from-cardekho

1. Objective

The objective of this assignment is to perform Exploratory Data Analysis (EDA) on a used-car dataset to understand its structure, identify data-quality issues, summarize important features, and discover patterns related to selling prices and vehicle characteristics.

The main objectives are to inspect the dataset dimensions and data types; identify missing values, duplicate records, and inconsistent formats; calculate descriptive statistics; visualize important variables; explore relationships between selling price and vehicle features; and summarize findings supported by the analysis.

The analysis demonstrates how data cleaning, statistics, and visualization can support understanding of a dataset. The results describe the available dataset and should not be interpreted as proof of causation.

2. Dataset Overview and Understanding

The Vehicle Dataset from CarDekho was obtained from Kaggle. It includes used-car information such as selling price, manufacturing year, kilometres driven, fuel type, transmission, ownership category, mileage, engine capacity, maximum power, torque, and number of seats.

Property	Details
Original dimensions	8,128 rows × 13 columns
Cleaned dimensions	6,926 rows × 14 columns
Feature types	Numerical and categorical
Derived feature	car_age (calculated as 2020 – year in the charted analysis)
Source	Kaggle: Vehicle Dataset from CarDekho

2.1 Description of columns

Column	Description
name	Car name or model
year	Manufacturing year
selling_price	Selling price of the car
km_driven	Distance travelled, in kilometres
fuel	Fuel category
seller_type	Seller category
transmission	Transmission type, such as Manual or Automatic
owner	Ownership category
mileage	Reported fuel efficiency
engine	Engine capacity, generally in CC
max_power	Maximum power, generally in BHP
torque	Reported engine torque
seats	Number of seats
car_age	Derived age feature: 2020 minus manufacturing year

3. Data Quality Assessment and Cleaning

The original data was inspected for missing values, duplicate records, and data types. Several specification columns contained missing values, and some numerical measurements were stored as text with units.

3.1 Initial missing-value counts

Column	Missing values before cleaning
mileage	221
engine	221
max_power	215
torque	222
seats	221

The initial dataset contained 1,202 duplicate records. Duplicate rows were removed using Pandas drop_duplicates(), reducing the row count from 8,128 to 6,926.

3.2 Cleaning steps

Duplicate removal: Duplicate rows were removed to reduce repeated records in the analysis.

Numerical conversion: Units were removed from mileage, engine, and max_power values, and the columns were converted to numeric data types. Invalid conversions were treated as missing values.

Missing-value handling: Missing values in mileage, engine, max_power, seats, and torque were filled with the median of their respective columns.

Torque processing: A numeric component was extracted from the torque text and converted to a numeric type. Original torque units were not standardized, so torque-related comparisons require caution.

Derived feature: car_age was calculated as 2020 minus year in the charted notebook output, which is consistent with the displayed age plot extending to approximately 37 years for a 1983 car.

3.3 Results after cleaning

Check	Recorded result
Original size	8,128 rows × 13 columns
Initial duplicate records	1,202
Cleaned size	6,926 rows × 14 columns
Missing values after cleaning	0 across all columns

The cleaning process made the dataset suitable for descriptive statistics and visual exploration. Median imputation may affect distributions, and the torque feature remains limited by inconsistent units. The final duplicate count after all processing was not available in the verified results and is therefore not reported.

4. Descriptive Statistical Analysis

Descriptive statistics were used to summarize the central tendency and range of numerical features. Frequency counts were used to identify common categories.

4.1 Numerical features

Feature	Mean	Median	Minimum	Maximum
year	2013.42	2014	1983	2020
selling_price	INR 517,271	INR 400,000	INR 29,999	INR 10,000,000
km_driven	73,996 km	70,000 km	1 km	2,360,457 km
mileage	19.46	19.44	0	42
engine	1,425.40 CC	1,248 CC	624 CC	3,604 CC
max_power	87.55 BHP	81.83 BHP	0	400 BHP
torque*	160.41	146	4.8	789
seats	5.42	5	2	14

*Torque values are numeric components extracted from mixed-format text; units were not standardized. Mileage is reported in the dataset's original units.

The mean selling price (about INR 5.17 lakh) is higher than the median (INR 4 lakh), consistent with a right-skewed distribution. Kilometres driven also varies substantially, with a maximum much higher than its median. The median number of seats is five.

4.2 Categorical features

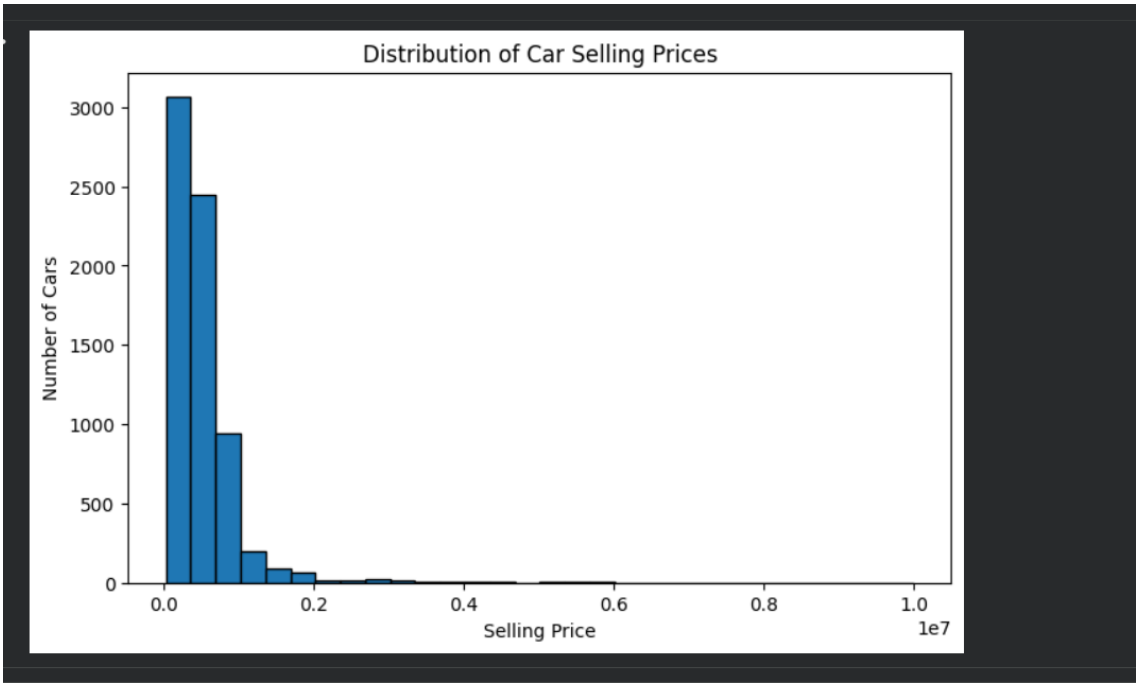
Feature	Most frequent category	Count
fuel	Diesel	3,755
seller_type	Individual	6,218
transmission	Manual	6,342
owner	First Owner	4,242
name	Maruti Swift Dzire VDI	118

The cleaned dataset contains 2,058 unique car names. Diesel, Individual seller, Manual transmission, and First Owner are the most frequent categories in their respective columns.

5. Exploratory Data Analysis and Visualizations

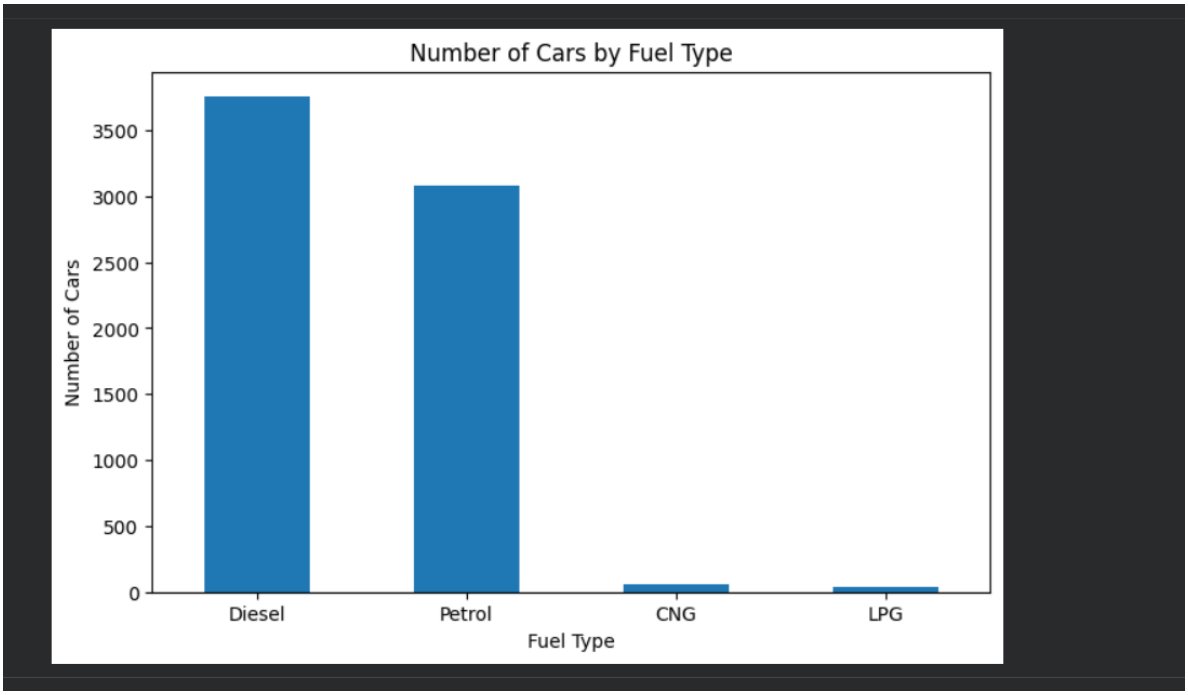
Six visualizations were included to examine distributions, category frequencies, group differences, age-price patterns, and correlations. Each figure is followed by an observation based on the available analysis.

5.1 Distribution of Car Selling Prices



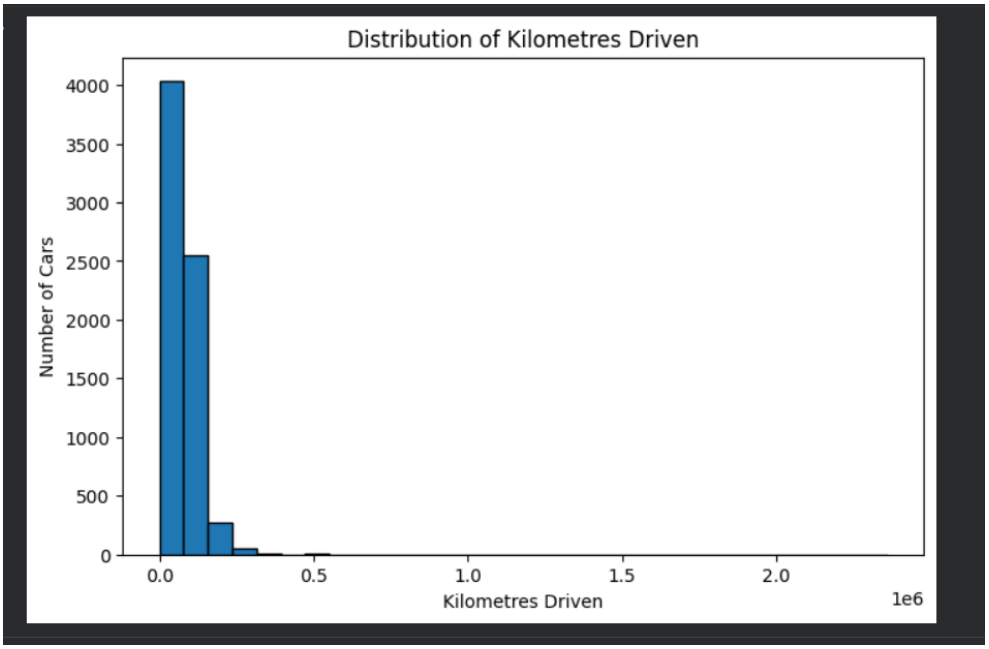
Observation: The histogram is right-skewed: most cars are concentrated in lower price ranges, while a smaller number of cars have much higher selling prices. This agrees with the mean selling price being higher than the median.

5.2 Number of Cars by Fuel Type



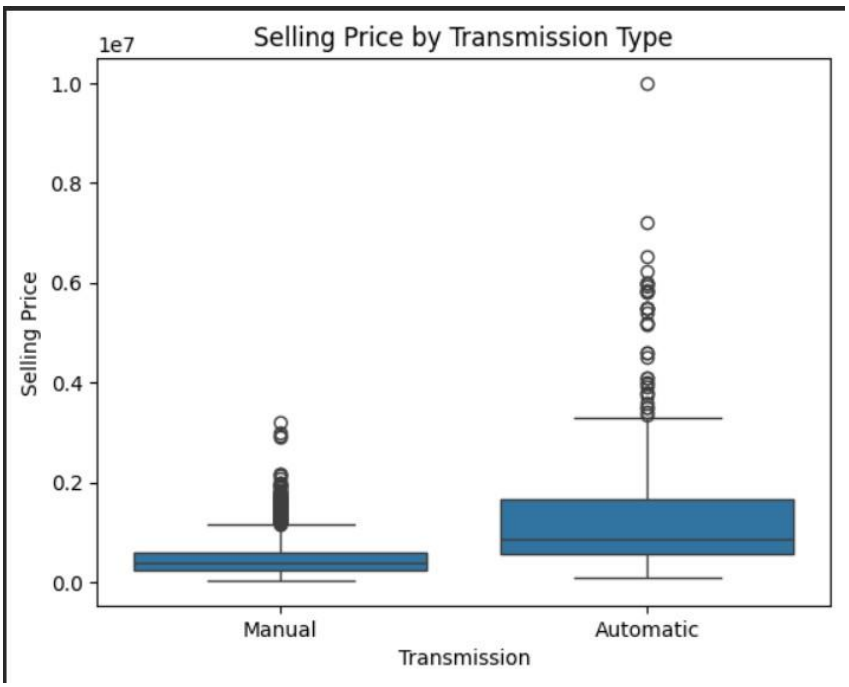
Observation: Diesel is the most frequent fuel category with 3,755 records, followed by Petrol. CNG and LPG appear much less frequently in the cleaned dataset.

5.3 Distribution of Kilometres Driven



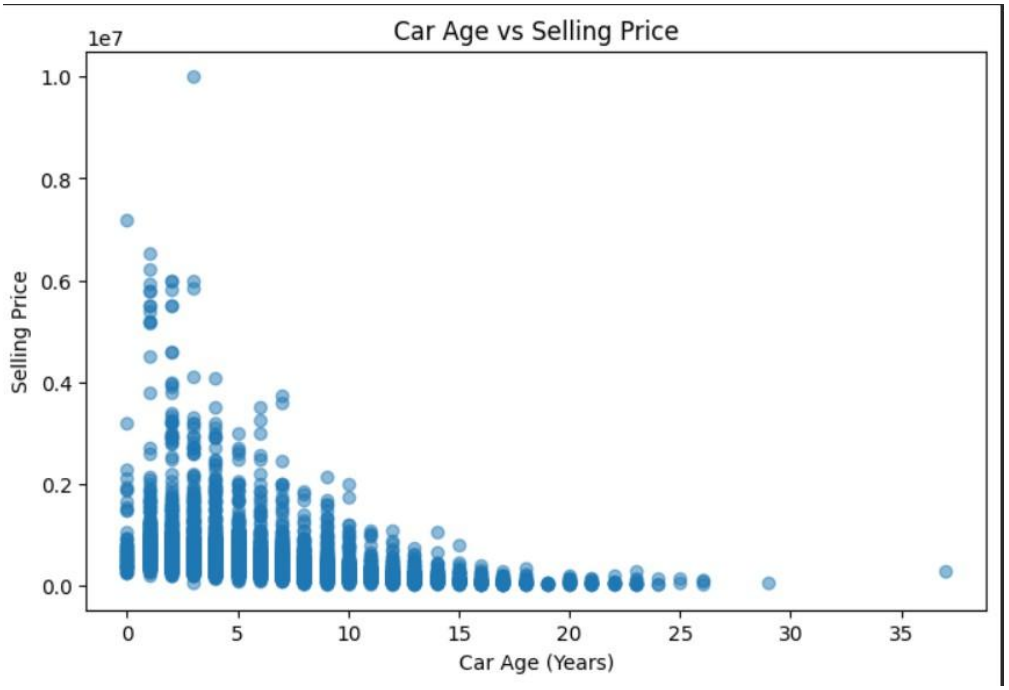
Observation: The histogram is strongly right-skewed. Most cars are concentrated at relatively low kilometre values, while a small number of cars have much higher recorded distances. The long right tail indicates a few very high-distance observations, consistent with the maximum recorded distance of 2,360,457 km.

5.4 Selling Price by Transmission Type



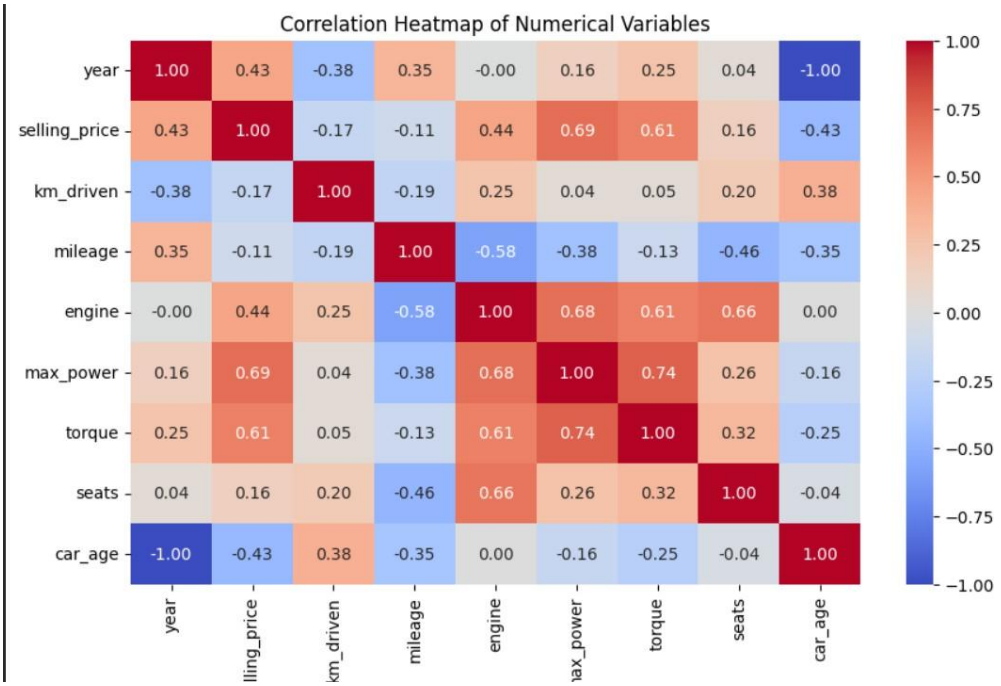
Observation: Automatic cars have a higher median selling price and a wider price distribution than manual cars in this dataset. Both categories contain high-price outliers, with particularly extreme values in the automatic category, reaching approximately INR1 crore.

5.5 Car Age vs Selling Price



Observation: The scatter plot shows a general downward pattern in selling prices as car age increases. Higher prices are concentrated mainly among newer cars, although prices vary considerably among cars of similar age.

5.6 Correlation Heatmap of Numerical Variables



Observation: The strongest positive correlation with selling price among the displayed numerical features is max_power (0.69), followed by torque (0.61), engine (0.44), and year (0.43). Selling price has weaker negative correlations with km_driven (–0.17) and mileage (–0.11).

Torque-related correlations require caution because the original torque units were not standardized. Correlation indicates association, not causation.

6. Key Findings and Insights

1. Selling-price distribution: The mean selling price is approximately INR 5.17 lakh, compared with a median of INR4 lakh. The histogram indicates a right-skewed distribution with a smaller number of expensive cars.

2. Fuel type: Diesel is the most frequent fuel category, with 3,755 records in the cleaned dataset.

3. Transmission: Automatic cars show a higher median selling price and a wider price distribution than manual cars in the displayed boxplot.

4. Car age: The scatter plot shows a general tendency for selling prices to be lower among older cars, though individual prices vary.

5. Numerical associations: Maximum power has the strongest positive correlation with selling price among the displayed numerical features (0.69). Torque, engine capacity, and manufacturing year also show positive correlations, subject to the torque-unit limitation.

6. Vehicle usage: Mean kilometres driven is about 73,996 km, and the maximum is 2,360,457 km. This wide range suggests that unusually high-distance records merit closer inspection.

7. Category frequencies: Individual sellers, manual transmissions, and first-owner cars are the most frequent categories in their respective columns.

These findings describe patterns in the analyzed dataset. They do not establish causal relationships and should not be generalized automatically to all used-car listings.

7. Limitations and Conclusion

7.1 Limitations

Missing-value imputation: Filling missing values with medians allows analysis to continue but may change feature distributions.

Extreme values: Selling prices and kilometres driven include extreme observations. These should be investigated before deciding whether they are errors.

Torque units: Torque values were extracted from mixed text formats without converting all measurements to a common unit, limiting comparability.

Correlation: Correlation measures linear association and does not prove that one feature causes a change in selling price.

Dataset coverage: The findings apply to the available dataset and may not represent the entire used-car market.

Derived car age: The charted analysis uses 2020 as the reference year. A different reference year would change the age values.

7.2 Conclusion

This assignment applied Exploratory Data Analysis to the Vehicle Dataset from CarDekho. The workflow included inspecting the dataset, removing duplicate rows, converting selected columns to numerical form, handling missing values with median imputation, and creating a derived car-age feature.

Descriptive statistics and six visualizations were used to study selling-price distribution, fuel-type frequency, kilometres driven, transmission-related price differences, car age, and numerical correlations. The analysis showed considerable variation in selling prices and associations between price and vehicle specifications. Maximum power had the strongest positive correlation with selling price among the displayed numerical features.